Student Performance Analysis Using R

MCA Mini Project Report

Prepared as a beginner-friendly R programming project.

1. Introduction

Student Performance Analysis Using R summarizes student marks and attendance. It calculates total and average marks, assigns a result and grade, identifies the top-performing student, and presents findings through charts.

2. Problem Statement

Manually reviewing marks for a class can take time and make comparisons difficult. This project uses R to organize records, calculate statistics, and visualize performance.

3. Objectives

- Calculate total and average marks for each student.
- Classify students as pass or fail using an illustrative 40-mark average threshold.
- Assign grades based on average marks.
- Identify the highest-performing student.
- Compare average marks across subjects.
- Visualize marks, pass/fail counts, and attendance against average marks.
- Export results to CSV files.

4. Tools and Technologies

Language: R; IDE: RStudio; data handling: base R data frames and functions; visualization: barplot(), pie(), and plot(); output: console reports and CSV files.

5. Dataset Description

The sample dataset contains 10 fictional students. Each record includes roll number, name, attendance, English marks, Mathematics marks, and Computer marks. Marks and attendance use a 0–100 scale.

6. Methodology

1. Create a data frame of sample student records.
2. Calculate total and average marks.
3. Assign Pass/Fail and grades using conditional statements.
4. Calculate class and subject summary statistics.
5. Find the student with the highest average.
6. Create bar charts, a pass/fail pie chart, and a scatter plot.
7. Export report tables as CSV files.

7. Expected Output

The Console displays the student report, class average, highest and lowest averages, pass/fail counts, top-performing student, and subject-wise averages. Charts appear in the Plots pane. Two CSV report files are saved in the working directory.

8. Conclusion

This project demonstrates basic educational data analysis using R, including data frames, arithmetic, conditional statements, summary functions, plotting, and CSV export. The results summarize class performance but are not a complete measure of student ability.

9. Limitations and Future Scope

The current version uses a small fictional dataset and simple grade thresholds. Future improvements could load real CSV/Excel data, handle missing marks, add a Shiny dashboard, compare semesters, and explore the relationship between attendance and marks.

10. Viva Questions (Short Answers)

What is R? R is a programming language used for statistical computing, data analysis, and visualization.

What is a data frame? A data frame is a table-like R structure whose columns can contain different types of data.

What does mean() do? It calculates the arithmetic average.

What does which.max() do? It returns the position of the maximum value.

What is ifelse()? It returns values depending on whether a condition is true or false.

What does table() do? It counts how many times each category occurs.

What does write.csv() do? It saves a data frame to a CSV file.

What is data visualization? It means presenting data using charts or graphs.

What is the purpose of a scatter plot here? It shows the relationship between attendance and average marks.

Are these real student records? No. They are fictional sample records created for demonstration.